Transducer Module Recognition in a Network Environment Based on IEEE 1451

Tércio A. dos Santos Filho[1], Antonio C. Oliveira-Jr[1], Dalton M. Tavares[1]
Marcos A. Batista[1] Alexandre C. R. da Silva[2] and Marcos N. Rabelo[3]
[1]Federal University of Goiás - UFG/CAC
Institute of Biotechnology
Catalão - Goiás - Brazil
e-mail: tercioas@gmail.com, antonio@catalao.ufg.br,
dmatsuo@gmail.com, marcos.batista@pq.cnpq.br
[2]São Paulo State University - UNESP
Department of Electrical Engineering - DEE
Ilha Solteira - São Paulo - Brazil
e-mail: acrsilva@dee.feis.unesp.br
[3]Federal University of Goiás
Graduate Program in Modeling and Optimization
Catalão, GO, Brazil
e-mail: rabelo@dmat.ufpe.br

Abstract— *The development of smart transducers networks demand a detailed study of processes to be controlled in order to achieve the user needs. This research proposes the development of networks with their own architectures in order to avoid compatibility issues at software or hardware level, which are common in proprietary communication protocols. The IEEE 1451 standard defines a set common commands and describes the features to NCAP and TIM to connect the transducers in network of way plug-and-play. This work presents a NCAP embedded using hardware and system operating dynamic implemented in DE2 kit. The NCAP has two interfaces based on IEEE 1451.5 (ZigBee) and IEEE 1451.2 (RS-232). Another important feature in this paper is the TEDS that were described of two ways, in STIM stored in no-volatile memory and to WTIM was used a text file stored in NCAP denominated virtual TEDS. The results were obtained through web page using a microcomputer to access the data in NCAP, like: temperature sensor, logs and the TEDS.*

Keywords: IEEE 1451, NCAP, RS-232, TIM, ZigBee.

1. Introduction

The transducers are components that convert a type of energy in other one, for example: electrical energy in mechanical (actuator) or physical in electrical (sensor). To realise the system data acquisition and control, the transducers are connected together with others devices, like: microcontroller and FPGA, denominated them smart transducers. The smart transducers network are developed using smart transducers module connected through a interface with others nodes or routers for realising the monitoring and/or control of the system in which being applied, these networks are denomineted DMC (Distributed Measurement and Control). The DMC systems development requires a great deal of engineering for its design, with the need to use tools and proprietary software, many with high costs for implementation. The transducer network can be applied in several environments in which wish monitoring or control, being used in smart houses, in industry, health and many others places, however, each manufacturer has a set of commands and protocols owner [1] [2].

To solve the problems of standardisation to NIST partnership with IEEE defined a set of standards and protocols for connecting smart transducers denomineted IEEE 1451 standard. The IEEE 1451 defines a set of protocols for wired and wireless distributed measurement, controls applications and plug-and-play through the TEDS (Transducer Electronic Data Sheet), in which is divided in two module: NCAP and TIM. The NCAP is a network node compost of hardware and software that provides the gateway functions between TIM and the user network, and the TIM is a module that contains signal conditioning, analog-to-digital or/and digital-to-analog conversion, frequency and digital converter and a interface to communicate with NCAP [3][4].

One technique to achieve plug-and-play was the definition of a minimum set of transducers and other optional features for more advanced functions. The TIM when is connected in NCAP it transfers TEDS information to protocol manager, in which performs the acknowledgment of transducers in the network automatically. each table in TEDS there is a format standardized based on IEEE 1451.0, being: Length, Data Block and Checksum, where, each line is denominated of TLV (Type/Length/Value). In Section 4.1 is described in more details the TEDS format.

The plug and play feature of smart transducers to its users and developers raises some advantages such as reduced time for parameterization of the system, advanced diagnostics, reduced time for repair and replenishment, advanced management of the hardware and automation of calibration [5].

For the implementation of TEDS, four blocks are required, Meta-TEDS TEDS TransducerChannel, User's Transducer Name TEDS TEDS and PHY. The other tables, Calibration TEDS, Frequency Response TEDS, Transfer Function TEDS, Text-based TEDS TEDS Commands, Identification TEDS, Geographic location TEDS TEDS extension Units, End User Application Specific TEDS TEDS defined and Manufacturer are optional. In this work, we used the TEDS mandatory for system testing and implementation of IEEE 1451.0-2007.

The NCAP described in this paper was developed over dynamic hardware and operating system embedded uClinux using C language based on IEEE 1451. It is important detach that the IEEE 1451.1 standard suggests object-oriented implementation, however, there is no requirement that actual implementation use of object-oriented implementation techniques [5]. The NCAP was made to be completely embedded and dynamic, both low level (hardware) like high level (software). To development in low level was used the SoPC Builder that there is default blocks described in VHDL or Verilog, and, in high level was configured and installed the operating system embedded uClinux. In Figure 1 presents the schematic of system.

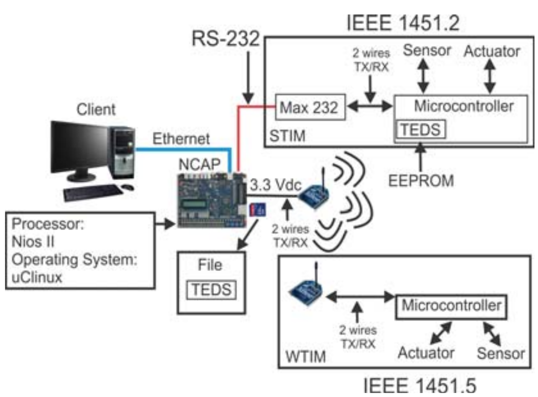

Fig. 1: Schematic of the implemented system.

The main propose of this paper is shows the TEDS of two ways, in TIM and in NCAP, and some features of sensor and actuator. The TEDS in TIM was stored using the memory flash of the Atmega8 microcontroller and in NCAP was stored at file. The TEDS stored in microcontroler describes the STIM developed with three transducers (temperature sensor and two step motors) and a RS232 interface to communication with NCAP. The TEDS stored at file in NCAP describes two transducers (temperature sensor and DC motor) and a wireless interface.

2. Related Works

The IEEE 1451 defines the TEDS description and communication between NCAP and TIM , involving a particular standard of the family and different technology approaches. Several implementations and applications of the IEEE 1451 smart transducer interface standards have been carried out using microprocessors, microcontroller, embedded commercial solutions and multicore technologies. Thus, many studies have been realized as: in [6] the TEDS format was described, and it was implemented a stand alone NCAP, using the RS-232 interface for realizing the communication with the TIM. The author also showed the system using a wireless communication based on IEEE 802.15.4. In [7] was described a NCAP developed in XML (eXtensible Markup Language) and RPC (Remote Procedure Call) with wireless interface IEEE 802.15.4 standard using a ZigBee module. The connection between NCAP and ZigBee module was made using the USB (Universal Serial Bus) standard.

In [8] described the IEEE 1451 and its main features, such as information model in which consists of a hierarchy of classes divided into three main categories: Block, Component and Service. The authors described the main blocks within each class (Block, Component and Service) and system tests, was made a software control water level in a tank. The authors also reported that, despite the advantages of the IEEE 1451.1 standard, there are still no commercial NCAPs available in the market.

In [9] was presented a prototype to the networks of mixed signal (Analogical/Digital) based on IEEE 1451.4 standard, in which was used a accelerometer sensor and was described the TEDS. The authors described the matter of a standardization to analogical transducers and the TEDS format. The authors showed a prototype developed in laboratory in which had a better understand about the standard and the integration between the IEEE 1451.4 and the others standards of IEEE 1451 family.

In [10], the author presented the development of an NCAP interfaces with two USB (Universal Serial Bus) implemented on a microcomputer. As the first step, the author provides a brief description of the IEEE 1451, each committee and when the standards were adopted. The second part presented the development of the system in which each TIM was implemented using the Freescale microcontroller (HC9S12DT256) and a memory (AT24C64A). The protocol used for communication between the NCAP and TIM was PTP (Precision Time Protocol) described by the IEEE 1588 standard in which the author describes the characteristics and size of messages.

In the work of [11] showed communication between the NCAP and WTIM, using the IEEE 1451.5, using the wireless interface based on IEEE 802.11 standard. The authors described the messages, commands, control patterns of the TIM, common commands, such as: TEDS reading, writing the TEDS, the TEDS request commands etc. And response commands to the NCAP, all represented by the class diagram. As a case study, the authors presented the recognition modules WTIMs and showed each step in recognition of the PHY TEDS in a graphical interface done in Java, validating recognition modules.

In other work [12] the author proposed the use of TEDS to store information concerning patient clinical history and diagnostic criteria in order to project learned and patient-adapting devices. The system uses the built-in information to optimize the data processing by adapting the diagnostic algorithm to the specific patient. The author presents the potential of TEDS in use of others applications.

3. Network Node Embedded

The NCAP is a network node with the function of getting data of external network, processing it and sending it to interface based on IEEE 1451.X. The NCAP is divided to hardware and software. The hardware is composed of processor, memory, I/O pins, driver to network and communication interface. The software is composted of an operating system, network protocols and transducers firmware. The operating system provides the logic interface between applications and hardware, and tools to realize the configurations necessary for the system. In Section 3.1 describes the hardware to development of network node and the Section 3.2 describes the operating system embedded to development of NCAP.

3.1 Hardware

The FPGA consists of logic cells arrangement, or configurable logic blocks, contained in a single integrated circuit. Each cell contains computational capacity to implement logic functions and perform routing for communication between blocks. The FPGA basically have logic blocks, blocks inbound and outbound keys and interconnection.

Among the families of commercially available FPGA by Altera, we have: Stratix II, Stratix, StratixGX, Cyclone II, APEX, APEX II, APEX 20K, Mercury, FLEX 10K, ACEX 1K, FLEX 6000 Devices and Excalibur [13].

To development of projects, the Altera developed a processor denominated Nios II that can to be described in VHDL or Verilog using the SoPC Builder. The SoPC Builder is a tool of the Quartus II environment from Altera, that there are blocks defined to implement in FPGA using the VHDL or Verilog language. The blocks provide the communication between the processor and the components, like: memory, I/O pins and interfaces, for example: UART and Ethernet [13].

The NCAP was made with two interfaces and with the Nios II/fast processor using 50MHz clock frequency internal and some components necessary to implement the embedded operating system, like: a timer block, Static Random Access Memory (SRAM) block, Syncronous Dynamic Random Acess Memory (SDRAM) block (for uClinux the minimum requirement is 8MB), Joint Test Action Group Universal Asynchronous Receiver/Transmitter - JTAG UART block, button block, SPI block (SD Card communication) and two Universal Asynchronous Receiver/Transmitter - UART blocks, used to communicate with WTIMs and STIMs.

3.2 Operating System Embedded - uClinux

The uClinux is a operating system embedded focused to work with devices without Memory Management Unit - MMU and offers support many processors, like: Coldfire, Axix, Etrax, ARM (Advanced RISC Machine), Atari 68k, Nios II and others distributed in the market. Today, uClinux is an operating system includes Linux kernel releases 2.0, 2.4 and 2.6 and user applications, libraries and toolchains [14]. The reconfiguration and recompilation of uClinux to each device is made using the software called uClinux-dist.

The uClinux-dist has configurations standards to build image with a set minimum resources predefined which can be modified or include others components [14]. In this paper the following applications were defined to create the image of the operating embedded system for NCAP: chmod, chown, date, df, echo, install, ls, mkdir, mv, pwd, clear, reset, vi, find, lsmod, modprobe, fdisk, arp, ftpget, ftpput, httpd, ifconfig, IP, ping, route, wget, free, kill, ps, uptime, msh and many others components available in the uClinux-dist.

The NCAP was implemented over operating system embedded uClinux. To communication were defined two interfaces in which the both owns the same features defined in the hardware and the operating system. The operating system was defined using the uClinux-dist software, that there are many options of tools to setup the system, as like: “Device drivers", “Character devices" and “Serial drivers". Select the options: “Altera JTAG UART support", “Altera JTAG UART console support" (if use a USB Blaster cable on a nios2-terminal), “Altera UART support", define “Maximum number of Altera UART ports", “baud rate" (in this project was defined 4800 bps) default baud rate for Altera UART ports and “bypass output" when no connection. Defined the interfaces and the applications was possible connect two TIMs (WTIM and STIM) and develop the NCAP using C language over operating system uClinux. For uClinux in the DE2 kit each serial port is defined by a special file in the /dev/ttySX, where, the dev represents the device directory and the ttySX (X port number) represents the access port. The ports defined in this project were ttyS0 (STIM) and ttyS1 (WTIM).

4. IEEE 1451.0 - TEDS

The IEEE 1451.0 - 2007 is a project that presents a common commands feature and TEDS family standard of smart transducers. This feature is independent of the physical medium of communication. This includes the basic functions required to control and manage smart transducers, protocols command and independence of the media format. The IEEE 1451.0 specifies the format of TEDS that are “tables” of data containing information of transducers (sensors/actuators) stored in nonvolatile memory inside the TIM. However, there are applications where storage in non-volatile memory is not practical for application, for example: when there is not memory in TIM, then the IEEE 1451.0 allows storage at other locations remotely calling them Virtual TEDS. The Virtual TEDS are electronic files that provide the same functionality implemented in memory of the TIM, however, they are not in TIM [5].

One technique to achieve feature plug-and-play was to define a minimum set of information from the transducers and other optional features for more advanced functions. The TIMs, to be connected to the NCAP, transfers data from the TEDS to manager protocol, which makes the recognition transducers network automatically, the system working of way plug-and-play.

The plug and play feature of smart transducers to its users and developers raises some advantages such as reduced time for parameterization of the system, advanced diagnostics, reduced time for repair and replacement, advanced management and automation hardware calibration [5].

For the implementation of TEDS, four tables are required, Meta-TEDS, TransducerChannel TEDS, User’s Transducer Name TEDS TEDS and PHY TEDS. The others tables, Calibration TEDS TEDS Frequency Response, Transfer Function TEDS, Text-based TEDS TEDS Commands, Identification TEDS, Geographic location TEDS TEDS extension Units, End User, Application Specific TEDS and Manufacturer TEDS are optionals. In this work was used the TEDS mandatory for system testing and implementation based on IEEE 1451.0-2007.

To test of feature plug-and-play, in this project was developed a module defined by IEEE 1451.2 denominated STIM using interface RS-232 and a module defined by IEEE 1451.5 denominated WTIM using interface ZigBee (point to point). The STIM there are three transducers, being: two step motors and one temperature sensor. The WTIM there are two transducers, being: temperature sensor and DC motor. To make the logic in both modules were used Atmega microcontroller.

4.1 TEDS Format

The TEDS format is common to any TEDS, where, the first field is the length and it is represented by 4 octets unsigned integer. The next block represents the TEDSt’s content, it can be represented by data binary or based on text. The last field in any TEDS is the checksum. The checksum is used to check the integrity of TEDSt’s data [5]. The Table 1 presents the structure TEDS general.

The TEDS fields are described like:

Table 1: Format generic to TEDS.

Field	Description	Type	Octet
—	TEDS length	UInt	4
1 to N	Data block	Variable	Variable
—	Checksum	UInt16	2

- **TEDS lenght -** it is the number of octets in the data block more two octets in the checksum;
- **Data block-** field that contains specific information according to each table TEDS. The fields that make up this structure are different for each type of TEDS and its structure is based on TLV, except the IEEE 1451.2-1997 and IEEE 1451.3-2003 [5]. The construction of each row within the table structure TEDS uses TLV defined as:
  - ***Type*** - field defined by 1 octet where this field represents the identification of TLV line;
  - ***Length*** - specific the number of octets in the field Value;
  - ***Value*** - content the informations of the TEDS field;
- **Checksum -** it is the complement of the sum of all octets preceded field including the initial size of the TEDS.

5. Transducer Interface Module - TIM

The TIM was developed using the microcontroller ATmega8 from Atmel, integrated circuit MAX 232 (wired) and a X-Bee Pro module (wireless) for interface between NCAP and TIM as demonstrated in Figure 1. To tests were used sensors and actuators, and the TEDS were described in EEPROM memory in the microcontroller like described in Figure 2. When the user connect the TIM in the NCAP, the TEDS are transfers to NCAP and made the recognition of the transducers connected to the TIM. Another way is when the TIM receives a command request, the TIM process the command and sends the reply message to the NCAP based on IEEE 1451.0 standard. The command can be data request from the sensor, actuator’s control or kind of TEDS.

6. TEDS Implementation in the Embedded Node

The NCAP software was developed in C language and using HTML CGI communication protocol. The IEEE 1451.1 standard suggest the implementation object orientation, however, does not demonstrate applications of structured programming. In this context, the work was developed using a new representation structure of NCAP, it being for each class suggested by the standard was represented by a file and each received a file name based on the IEEE 1451.1 standard, such as: IEEE 1451_<name file>.

Fig. 2: TEDS description in EEPROM memory of ATmega8.

In “TEDS Descriptions” option presents which TEDS that are stored on network node NCAP. In this paper was described the four tables required for each TIM, where each TEDS table was stored in differents files and specified based on communication interface to which it belongs. As an example of description of the file name, have, RS232_meta_TEDS, which sets the RS-232 interface and Meta-TEDS TEDS table describes what is being described.

The NCAP recognizes the TIM when both connects and transfers the UUID of the TIM to NCAP, if the identifier is not present in NCAP is done reading the TEDS and stored in a separate file. However, if the identifier UUID is present in NCAP, then the reading is performed in the TEDS itself, that is done for recognition. In Figure 3 presents a user interface for selecting modules reading the TEDS to show to user.

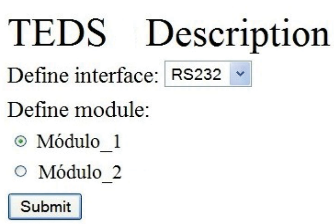

Fig. 3: Interface selection reading the TEDS.

In Figure 4 shows an example of reading of the TEDS in hexadecimal.

In Figure 5 presents an example of features relating to the STIM1-RS232 module.

In Figure 6 presents the description TEDS for step motor.

In Figure 7 presents the description TEDS for interface.

Figure 8 shows the system implemented in the laboratory with the NCAP connected to microcomputer through the Ethernet network and the two TIMs (STIM and WTIM) connected to the NCAP using the wireless and wired network.

Meta TEDS

Field Type	Field Name	Description	Data Type	Lenght	Value
---	Lenght	Octets Numbers	UInt32	4	0 0 0 25
3	TEDSID	Identification TEDS	Uint32	4	0 1 1 1
4	UUID	Universal Unique Identification	UUID	10	8 FB 61 B4 80 81 F6 43 A1 B1
A	OHoldOff	Time Limit to response	Float32	4	40 A0 0 0
C	TestTime	Time to self test	Float32	4	40 0 0 0
D	MaxChan	Transducer channel number	UInt16	2	0 3
----	Checksum	----	UInt16	2	6 2F

Fig. 4: Example of reading the TEDS stored on file in NCAP.

Transducer Module

Module Identification:	8 FB 61 B4 80 81 F6 43 A1 B1
Time Limit to response	5.0000
Time to self test	2.0000
Transducer channel number	3

Transducer - Channel 1

Calibration Key	0
Channel Type	Sensor
Physical Units	32 1 0 39 1 82
Low Limit	0.0000
High Limit	55.0000
Operational Error	0.5000
Self Test	1
Sample	28 1 0 29 1 1 30 1 8
TransducerChannel update time	0.1000
TranducerChannel read setup time	0.0000

Fig. 5: Description of TEDS for the STIM1_RS232 module.

7. Conclusion

In this paper has presented the implementation of an embedded NCAP to access the TIMs through wireless and wired interfaces. The NCAP was fully implemented in the DE2 kit, using the Nios II soft-core and the uClinux embedded operating system. For this project two interfaces, wired and wireless have been considered to test the system and to implement the IEEE 1451 concepts, by means of a WTIM and a STIM module. With the implemented uClinux operating system both interfaces can be configured.

Another important aspect was the TEDS described using the file in the NCAP and it stored in TIM using the micro-controller in the flash memory, in which demonstrated the same functionality to both implementation and it becoming the TIMs plug-and-play. This approach allows introducing relevant aspects in the creation of digital systems such as code reuse and technological independence.

Acknowledgment

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Transducer - Channel 2

Calibration Key	0
Channel Type	Atuador
Physical Units	32 1 0 33 1 82
Low Limit	0.0000
High Limit	360.0000
Operational Error	1.8000
Self Test	0
Sample	28 1 0 29 1 1 30 1 8
TransducerChannel update time	0.1000
TranducerChannel read setup time	0.0000
TransducerChannel sampling period	0.1000
TransducerChannel warm-up time	41 F0 0 0
TransducerChannel self-test time requirement	30.0000
Sampling	31 1 0
End-of-data-set operation attribute	1
Data Transmission Attribute	1
Actuator-halt attribute	1

Fig. 6: Description of TEDS for step motor.

PHY TEDS

Field Type	Field Name	Description	Data Type	Lenght	Value
---	Lenght	N de octetos na META TEDS	UInt32	4	0 0 0 19
3	TEDSID	TEDS Identification Header	UInt32	4	0 C 1 1
10	RS232	IEEE 1451.2 - RS232 Physical type	UInt8	1	1
12	MaxBPS	Max data throughput	UInt32	4	0 0 4 B0
13	MaxCDev	Max Connected Devices	UInt16	2	0 1
14	Encrypt	Encryption	UInt16	2	0 0
15	Authent	Authentication	Boolean	1	0
16	MinKeyL	Min Key Length	UInt16	2	0 0
17	MaxKeyL	Max Key Length	UInt16	2	0 0
18	MaxSDU	Max SDU Size	UInt16	2	0 1
19	MinALat	Min Access Latence	UInt32	4	0 0 0 5
20	MinTLat	Min Transmit Latency	UInt32	4	0 0 0 5
21	MaxXact	Max Simultaneous Transactions	UInt8	1	1
22	Battery	Device is battery powered	UInt8	1	1

Fig. 7: Description of TEDS for the STIM1_RS232 module.

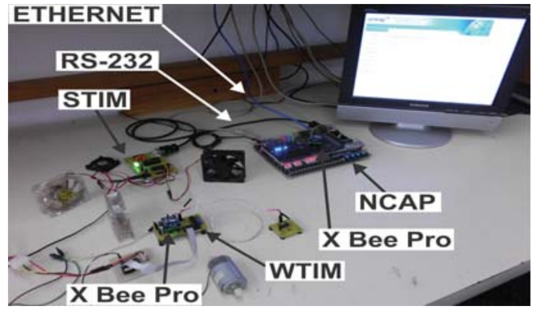

Fig. 8: Picture of the system implemented in laboratory.

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